Day 6 – Prefix Sum & Subarray Problems

1️⃣ Prefix Sum Concept

Prefix Sum kya hota hai?

Prefix sum ka matlab hai array ke elements ka cumulative sum.

Matlab:

prefix[i] = arr[0] + arr[1] + arr[2] + ... + arr[i]

Example

Array:

arr = [2, 4, 6, 8]

Prefix Sum Array:

Index	Calculation	Prefix
0	2	2
1	2+4	6
2	2+4+6	12
3	2+4+6+8	20

Result:

prefix = [2, 6, 12, 20]

Java Code – Prefix Sum Array

```java
public class PrefixSumExample {

    public static void main(String[] args) {

        int[] arr = {2,4,6,8};
        int n = arr.length;

        int[] prefix = new int[n];

        prefix[0] = arr[0];

        for(int i = 1; i < n; i++){
            prefix[i] = prefix[i-1] + arr[i];
        }

        for(int i = 0; i < n; i++){
            System.out.print(prefix[i] + " ");
        }

    }

}
```

Output

2 6 12 20

Time Complexity → O(n)
Space Complexity → O(n)

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2 Prefix Sum ka Use

Prefix sum ka use hota hai subarray sum quickly calculate karne me.

Formula:

Sum(L to R) = prefix[R] - prefix[L-1]

Example:

arr = [2,4,6,8]
Find sum from index 1 to 3

prefix[3] - prefix[0]
20 - 2 = 18

3 Subarray Problem

Subarray kya hota hai?

Array ke continuous elements ka group.

Example:

arr = [1,2,3]

Possible subarrays:

[1]
[2]
[3]
[1,2]
[2,3]
[1,2,3]

---

4 Find Sum of All Subarrays

Java Code

```
public class SubarraySum {

    public static void main(String[] args) {

        int[] arr = {1,2,3};
        int n = arr.length;

        for(int i = 0; i < n; i++){

            int sum = 0;

            for(int j = i; j < n; j++){

                sum = sum + arr[j];

                System.out.println("Subarray sum = " + sum);
            }
        }

    }

}
```

Output

1
3
6
2
5
3

Time Complexity → $O(n^2)$

5 Subarray Sum Using Prefix Sum

```
public class SubarraySumPrefix {

    public static void main(String[] args) {

        int[] arr = {1,2,3,4};
        int n = arr.length;

        int[] prefix = new int[n];

        prefix[0] = arr[0];

        for(int i = 1; i < n; i++){
            prefix[i] = prefix[i-1] + arr[i];
        }

        int L = 1;
        int R = 3;

        int sum;

        if(L == 0){
            sum = prefix[R];
        }else{
            sum = prefix[R] - prefix[L-1];
        }

        System.out.println("Subarray sum = " + sum);

    }

}
```

Output

Subarray sum = 9

Problem 1 – L se R tak Sum Find Karo

Example

Array = [2,4,6,8]

L = 1

R = 3

Java Code

```
public class RangeSum {

    public static void main(String[] args) {

        int[] arr = {2,4,6,8};
        int n = arr.length;

        int L = 1;
        int R = 3;

        int[] prefix = new int[n];

        prefix[0] = arr[0];

        for(int i = 1; i < n; i++){
            prefix[i] = prefix[i-1] + arr[i];
        }

        int sum;

        if(L == 0){
            sum = prefix[R];
        } else {
            sum = prefix[R] - prefix[L-1];
        }

        System.out.println("Sum from index " + L + " to " + R + " = " + sum);

    }

}
```

Output

Sum from index 1 to 3 = 18

Problem 2 – Total Number of Subarrays

Formula:

Total Subarrays = n(n+1)/2

Java Code

```
public class TotalSubarrays {

    public static void main(String[] args) {

        int[] arr = {1,2,3,4};
        int n = arr.length;

        int total = n * (n + 1) / 2;

        System.out.println("Total number of subarrays = " + total);

    }

}
```

Output

Total number of subarrays = 10